Week of 31st

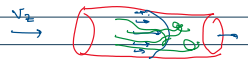
Monday, 30 October 2023 4:51 PM

Realistic & practical flow:

Turbulent flow: can use the same equations (microscopic balances)

↳ However, flow is no longer unidirectional & SS } time dependent and 3D (PDE)

($v_r, v_\theta \neq 0$)



↓

Cannot solve for velocity profile!

↓

Use experiments to fill in the gaps! → {Empirical correlations}

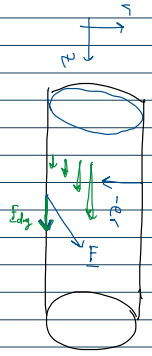
{Pressure drop (F_{drag}) vs. flow rate (Q) relationships}

Fluid force on a surface S: $\underline{F} = \iint_S [\hat{n} \cdot \underline{\Pi}]\big|_{@surface} \, dS$

This force is being exerted by the fluid on the surface

↓

This is the cause of pressure drop in pipe flow!!



$\underline{F}$ ⇒ total force

$\underline{F}_{drag} = \underline{F}_z$ (i.e. z^{th} component of force)

$$\therefore \underline{F}_{drag} = \int_0^L \int_0^{2\pi} \hat{e}_z \cdot \left(-\hat{e}_r \cdot \underline{\Pi}\big|_R\right) R \, d\theta \, dz$$

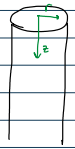
↓ non-dimensionalize

$$\underline{F}_{drag} = \int_0^L \int_0^{2\pi} -\mu \left(\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z} \right)\Big|_R R \, d\theta \, dz$$

Fluid drag in turbulent flow on a piper surface of unit normal $-\hat{e}_r$

Practice Questions

Laminar flow in a pipe: (Unidirectional, SS) } Continuity + Navier → BC → v profile



$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{r\theta z}$; $\underline{g} = \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}_{r\theta z}$ $\quad p = p(z)$

↓

+ Pressure profile: $p(z) = -\left(\frac{p_0 - p_L}{L}\right) z + p_0$

+ Velocity profile: $v_z(r) = \frac{(p_0 - p_L + \rho g L) R^2}{4 \mu L} \left[1 - \left(\frac{r}{R}\right)^2 \right]$

$$\underline{\underline{\Pi}}(r,z) = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$$= -p\underline{\underline{I}} + \begin{pmatrix} 0 & 0 & \mu\frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \mu\frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix}_{rθz}$$

$$= \begin{pmatrix} -p(z) & 0 & (p_L - p_0)\frac{r}{2L} \\ 0 & -p(z) & 0 \\ (p_L - p_0)\frac{r}{2L} & 0 & -p(z) \end{pmatrix}$$

→ Neglect gravity:

$$v_z(r) = \frac{(p_0 - p_L + \rho g L)R^2}{4\mu L}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$= \frac{(p_0 - p_L)R^2}{4\mu L} \cdot \frac{2}{2}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$v_z(r) = 2\langle v\rangle\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$\frac{\partial v_z}{\partial r} = \frac{-4\langle v\rangle r}{R^2}$$

$$\underline{F}_{drag} = \int_0^L \int_0^{2\pi} -\mu\left(\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z}\right)\Big|_R R\,d\theta\,dz$$

$$= \int_0^L \int_0^{2\pi} -\mu\left(\frac{-4\langle v\rangle r}{R^2}\right)\Big|_R R\,d\theta\,dz$$

$$= (4\mu\langle v\rangle)(2\pi \cdot L)$$

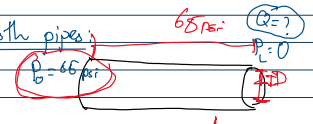
$$= 4\mu 2\pi L \cdot \left(\frac{(p_0 - p_L)R^2}{8\mu L}\right)$$

$$\boxed{\underline{F}_{drag} = \pi R^2 (p_0 - p_L)}$$

$$\Rightarrow Q = \langle v\rangle \pi R^2 = \frac{\pi (p_0 - p_L) R^4}{8\mu L}$$

$$\langle v\rangle = \frac{(p_0 - p_L)R^2}{8\mu L}$$

Turbulent flow in smooth pipes:



Prandtl correlation: $\frac{1}{\sqrt{f}} = 4\log\left[Re\sqrt{f}\right] - 0.4$ (Smooth pipes) 7.156

$Re = \frac{\rho\langle v\rangle D}{\mu}$

$f = \frac{\Delta p D}{2L\langle v\rangle^2}$ (Pipe friction factor from experiments)

Algorithm 1:

1) Guess Q_{guess} I will give $\langle v\rangle$ ($\langle v\rangle = \frac{Q}{A_c}$)
2) Find Re
3) Find f (Prandtl correlation)
4) Find Δp ($\rightarrow \Delta P$)

friction factor

Iterative solver:
+ guess of f
+ $g(f) = 4\log\left[Re\sqrt{f}\right] - 0.4$
+ New f: $f_{new} = \frac{1}{(g(f))^2}$
+ Confirm / compare f w/ f_{new}

$\frac{1}{\sqrt{f}} = g(f)$ → $\sqrt{f} = \frac{1}{g(f)}$ → $f = \frac{1}{(g(f))^2}$

$\Rightarrow 4\log\left[\frac{\rho\langle v\rangle D}{\mu}\sqrt{\frac{\Delta p D}{2L\rho\langle v\rangle^2}}\right] - 0.4 = \frac{1}{\sqrt{f}}$

$\Rightarrow 4\log\left[\frac{\rho D}{\mu}\sqrt{\frac{\Delta p D}{2L\rho}}\right] - 0.4 = \frac{1}{\sqrt{f}}$

2) Guess Q_{guess} (will give $\langle v \rangle$)
3) Find Re
4) Find f_1 — friction factor — Prandtl correlation
5) Find Δp_1
6) Calculate Δp_2 = known value − Δp_1
7) Solve for "LHS" (given Δp_2) → gives f_2 → $\langle v \rangle$ (friction factor) using
8) Solve for Q_{calc} [$Q_{calc} = \langle v \rangle \pi R^2$]
9) Compare this to step 1 Q_{guess} → if close then stop
↳ else iterate

Q_{new}

$$\Rightarrow 4 \log \left[\frac{\rho \langle v \rangle D}{\mu} \sqrt{\frac{\Delta p D}{2 L \rho \langle v \rangle^2}} \right] - 0.4 = \frac{1}{\sqrt{f}}$$

$$\Rightarrow 4 \log \left[\frac{\rho D}{\mu} \sqrt{\frac{\Delta p D}{2 L \rho}} \right] - 0.4 = \frac{1}{\sqrt{f}}$$

$$\Rightarrow 4 \log \left[\sqrt{\frac{\rho \Delta p D^3}{2 L \mu^2}} \right] - 0.4 = \frac{1}{\sqrt{f}}$$

"LHS"

↳ $f(v)$

Turbulent flow in Connected smooth pipes

$\Delta p_2 = p_2 - p_3 = p_2 - p_4$

1) Guess Δp_2 — "LHS"
find f — friction factor
$\langle v \rangle$
Q_1 Q_2

$\langle v_1 \rangle, \langle v_2 \rangle$, $\Delta p_2 \rightarrow Q_1, Q_2$

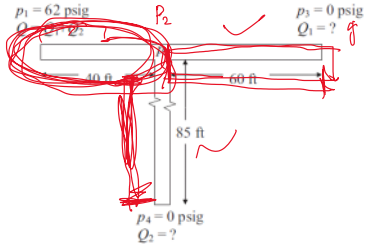
2) Total Q ($Q_1 + Q_2 = Q$)

Now use Algorithm 1 ! (until step 5)

↳ $\Delta p_1 = p_1 - p_2$

3) $(\Delta p_2)_{calc} = 62 - \Delta p_1$

4) Iterate until $(\Delta p_2)_{calc} \approx (\Delta p_2)_{guess}$



EXAMPLE 7.9. *A 40-foot section of 1.0-inch ID piping branches into two pipes of the same diameter, one of which is 60.0 feet long and one of which is 85.0 feet long (Figure 7.23). The main pipe is connected to the municipal water supply, which supplies a constant water pressure of 62 psig at the pipe entrance. What are the flow rates through the two pipe exits? What is the pressure at the splitting point? Assume smooth pipes.*

① Since we don't know Q
↳ we first need to solve for Q_1 & Q_2 by guessing Δp_2 value.

② Once we know Q → then focus on small initial 40 ft pipe and calculate Δp_2 value independtly
↳ See if it matches?
↳ if not we iterate!

Turbulent flow in connected rough pipes

Colebrook correlation: $\frac{1}{\sqrt{f}} = -4 \log \left(\frac{\varepsilon}{D} + \frac{4.67}{Re \sqrt{f}} \right) + 2.28$ → Prandtl correlation

$g(f)$

$$f = \frac{\Delta p D}{2 L \rho \langle v \rangle^2} \quad ; \quad Re = \frac{\rho \langle v \rangle D}{\mu}$$

$$\frac{1}{\sqrt{f}} = -4 \log \left(\frac{\varepsilon}{D} + \frac{4.67}{\left(\sqrt{\frac{\rho \Delta p D^3}{2 L \mu^2}} \right)} \right) + 2.28$$

"LHS" (when we only know Pressure drop)

"LHS" (when we only know Pressure drop & no average velocity)